MILITARY SPECIFICATION

PROJECTILE, 8 INCH: HE: M509A1 LOADING, ASSEMBLING AND PACKING

This specification is approved for use within US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of projectile designated as Projectile, 8 Inch: HE, M509E1.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The folllowing specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-P-48099	- Propellant, M10 (for use in Special Application)
MIL-P-48395	- Pellet, Explosive Composition (for use in Ammunition)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-101	- Preservatoin, Packaging and Packing Materials, Test Procedures

MILITARY

MIL-STD-286	- Propellants, Solid; Sampling, Examination and Testing
MIL-STD-453	- Inspection, Radiographic

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.9)

US ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKING DRAWINGS

9229038	- Pallet for Projectile, 8 INch: He, M509E1 Empty and Loaded
9236500	- Fuze, Mechanical Time and Superquick M577
9236502	- Plug Assembly
9236504	- Lead, Multi-Purpose
9272016	- Charge, Spotting, Projectile
9362612	Projectile, 8-Inch: HE, M509A1

INSPECTION EQUIPMENT DRAWINGS

D8841732	- Test Fixture
A9202990	- Operating Instructions

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications - The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents wich is current on the date of the soliciation.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts, 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 DFR 100-199 (latest revision)').

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Method E-18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

2.2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts - The projectile assembly and all components thereof shall comply with all requirements specified on Drawing (Dwg.) F-9362612, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Strength of heat seal. The specimen shall withstand application of a static load of 2.5 pounds per lineal inch, minimum without peel-back of the heat seal and without tensile failure of the area adjacent to the heat seal.

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3.4 First Article Inspection - This specification contains technical provisions for first article inspection. Requirements for the submission of first article saples by the contractor shall be as specified in the contract.

3.5 Static functioning test of booster PA 530. The booster PA 530 (Dwg 9378247) shall function high order as demonstrated by initiation of the adjacent body loading assemblies.

3.6 Ballistic functioning tests -

3.6.1 Projectile - The projectile shall function satisfactorily (See 6.6).

3.6.2 Grenade - The grenade shall function satisfactorily upon ejection from the projectile (See 6.6).

3.6.3 Safety - There shall be no premature burst (See 6.6.1) or evidence thereof, and there shall be no separation of metal parts either in the bore of the weapon or in flight, (prior to initiation of the time fuze), in any of the ballistic tests, including those rounds fired in the safety phase.

3.7 Radiographic examination of projectiles - Prior to forwarding the projectiles to the proving ground for ballistic testing, the projectiles shall be subjected to radiographic examination. The projectiles shall be examined for the following:

a. Gap between base and adapter (see Note)
b. Gap between adapter and grenade (see Note)
c. Excessive gap at take-up pad area between grenade and forward plate (see Note)
d. Gap between grenades (see Note)
e. Safety pin or safety clip shall not be present
f. Any missing or improperly positioned component or assembly normally discernible on the radiograph.

NOTE: Perceptible evidence of gap (in excess of 0.015 inches) shall be considered a reject for this inspection.

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3.8 Workmanship - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspectio of requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of Inspections - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection -

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4.3.1 Submission - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall comsist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Projectile, 8 Inch:		
HE, M509A1	9362612	10
Expulsion Charge Assembly	9332505	25
Booster PA 530	9378247	25
Shim	9272006	25
Pellet	9378249	15
Obturator	9287983	5
Lid, Container, Expulsion Charge	12576778	See Note
Cup, Container, Expulsion Charge	12576779	See Note
Container Assembly, Expulsion Charge	12576777	50
Expulsion Charge Assembly	12576776	25

Note: Five (5) samples from each mold cavity

4.3.2 Inspections to be Performed - See MIL-A-48078 and Table I herein.

4.3.3 Rejection - See MIL-A-48078. In addition, the first article sample shall be rejected if the ballistic sample is rejected (see 4.4.3.2.3(. The projectile replacement provisions of 4.4.3.2.4 and the retest provisions of 4.4.3.2.5 shall also apply to the first article ballistic functioning tests.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48091G (AR)

PARAGRAPH	TITLE			SHEET 1 OF 2	DRAWING NUMBER
	Projectile, 8 Inch: HE, M509A1				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 8 Inch: HE, M509A1 (9362612)				
	Examination for defects	10	-	3.2	4.4.2.15
	Ballistic Functioning at Zone 1	3	-	3.6	4.5.7.1
	Ballistic Functioning at Zone 9	3	-	3.6	4.5.7.2
	Ballistic Functioning at Zone 9X - Safety Phase	3	-	3.6	4.5.7.3
	Contingency round for replacement or retest	1	-	-	4.4.3.2.4
	Radiographic examination of projectile for engineering information only	10	-	3.7	4.5.8
	Expulsion Charge Assembly (9332505)				
	Examination for defects	25	-	3.2	4.4.2.4
	Booster PA 530 (9378247)				
	Examination for defects	25	-	3.2	4.4.2.6
	Shim (9272006)				
	Examination for defects	25	-	3.2	4.4.2.18
	Pellet (9378249)				
	Examination for defects	15	-	3.2	4.4.2.19
*	Obturator (9287983)				
	Examination for defects	5	-	3.2	4.4.2.23
	Breaking load test	2	-	3.2	4.5.15

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Projectile, 8 Inch: HE, M509A1		SHEET 2 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
*	Lid, Containeer, Expulsion Charge (12576778) Examination for defects	(a)	-	3.2	4.4.2.24
*	Cup, Container, Expulsion Charge (12576779) Examination for defects	(a)	-	3.2	4.4.2.25
*	Container Assembly, Expulsion Charge (12576777) Examination for defects	50	-	3.2	4.4.2.27
*	Expulsion Charge Assembly (12576776) Examination for defects	25	-	3.2	4.4.2.28

NOTES: (a) Five (5) samples from each mold cavity.

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation - Inspection lots shall xomply with the lot formation provisions of MIL-A-48078. In addition inspection lots of Projectiles shall contain:

a. Grenades from one finally accepted lot interfix number from one manufacturer (See Note 1).

b. Projectile body assemblies from one lot interfix number from one manufacturer.

c. Bases from one lot interfix number from one manufacturer.

d. Forward plates from one lot interfix number from one manufacturer.

e. Adapters from one lot interfix number from one manufacturer.

f. Expulsion charge propellant from one manufacturer and no more than two serial lot numbers.

g. Expulsioin charge assemblies from one lot interfix number from one manufacturer.

h. Booster PA 530 pellet charge material from one manufacturer and no more than two serial lot numbers.

i. Spacers from one lot interfix number from one manufacturer.

j. Spacers, small from one lot interfix number from one manufacturer.

k. Spacers, large from one lot interfix number from one manufacturer.

l. Keys from one lot interfix from one manufacturer.

m. Loaded projectiles of one weight zone (See Note 2).

* n. Lids, expulsion charge container, from one lot interfix number from one manufacturer.

* o. Cup, expulsion charge container, from one lot interfix number from one manufacturer.

NOTE 1: Only those grenade lots which have satisfactorily passed the grenade functioning test shall be assembled into the projectile.

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NOTE 2: Projectiles falling out of the inspection lot criteria due to the zone weight restriction may be accumulated into a hydrid lot provided they originally met all other lotting restrictions during manufacture and the original lot passed all acceptance tests. Further ballistic testing of this hybrid lot is not required.

4.4.2 Examination - See MIL-A-48078.

a. Sampling Plans - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be utilized if approved by the Procuring Activity.

b. Automatic gage. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

c. Alternate inspection procedure. The contractor may propose the use of alternate inspection procedures in lieu of conducting sampling inspection for defects listed in this specification (see 6.12).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Bag, Charge	1 OF 1	9332507
			NEXT HIGHER ASSEMBLY: 9332506

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of bag		0.040%	3.2	Gage
102	Distance to end heat seal max.		0.40%	3.2	Gage
103	Distance to side heat seal		0.40%	3.2	Gage
104	Minimum width of both heat seals over entire length		0.40%	3.2	Gage
105	Heat seal incomplete (2 places)		0.40%	3.2	Visual
106	Strength of heat seal	(a)		3.3	4.5.11
107	Location of eyelet		0.40%	3.2	Gage
108	Location of end stitching		0.40%	3.2	Gage
109	Distance to side stitching		0.40%	3.2	Gage
Minor					
201	Stitching incomplete (2 places)		0.65%	3.2	Visual
202	Fold width eyelet end		0.65%	3.2	Gage
203	Fold misalignment max		0.65%	3.2	Gage
204	Thickness of assembled eyelet		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) Sample 20 per lot, reject on one (1) or more defectives.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9282022
4.4.2.2	Cover, Expulsion Charge				NEXT HIGHER ASSEMBLY 9332506
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of rim		0.40%	3.2	Gage
102	Thickness of rim, minimum		0.40%	3.2	Gage
103	Total height		0.40%	3.2	Gage
104	Angle between rim and angled portion		0.40%	3.2	Gage
105	Thickness of angled portion, minimum		0.40%	3.2	Gage
106	Height of rim		0.40%	3.2	Gage
107	Location of tabs		0.40%	3.2	Gage
108	Distance bwtween tab and angled portion (4 places)		0.40%	3.2	Gage
109	Radius missing		0.40%	3.2	Visual
110	Crack or puncture		0.40%	3.2	Visual
111	Burr		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Charge Loading Assembly	1 OF 1	9332506
			NEXT HIGHER ASSEMBLY 9332505

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined				
Major					
101	Pull wire or bag loose with respect to cover		0.40%	3.2	Visual/Manual
102	Cover damaged		0.40%	3.2	Visual
103	Clinch of stud improper		0.40%	3.2	Visual
104	Bag damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.4	Expulsion Charge Assembly	1 1	9332505
			NEXT HIGHER ASSEMBLY 9362612

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak test	(a)		3.2	4.5.2
102	Weight of propellant charge incorrect		0.40%	3.2	Gage
103	Bag seal incomplete (3 places)		0.40%	3.2	Visual
104	Moisture and total volatiles content of propellant	(a)	0.40%	3.2	4.5.1
105	Bag damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample in accordance with the applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Booster PA 530 - Prior to Crimping of Closing Disc Assy		9378247
			NEXT HIGHER ASSEMBLY 9378247

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cup cracked or damaged		0.40%	3.2	Visual
102	Pellet cracked or damaged		0.40%	3.2	Visual
103	Excessive adhesive at interface (if applicable)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Booster PA 530 - After Crimping of Explosive Load Assy		9378247
			NEXT HIGHER ASSEMBLY 9362612

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Closing disc damaged		0.40%	3.2	Visual
102	Closing disc bulge not exceeding man. length of booster assy		0.40%	3.2	Visual
103	Diameter		0.40%	3.2	Gage
104	Length		0.40%	3.2	Gage
105	Cracks in aluminum crimp area	15	*	3.5	4.5.5
106	Static functioning test		0.65%	3.2	4.5.5
107	Loose pellets in booster assy				Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

* No rejects allowed.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Assembly, Forward Plate and O-Ring				None
					NEXT HIGHER ASSEMBLY 9362612
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	O-ring not greased		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Projectile Metal Parts Assembly - Prior to Any Loading	1 OF 1	10551896
			NEXT HIGHER ASSEMBLY: 9362612

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lubricating grease not applied to interior surface of body		0.40%	3.2	Visual
102	Presence of foreign material on inside of body		0.40%	3.2	Visual
103	Forward plate, packing, or forward plate pad, missing		0.40%	3.2	Visual
104	Lubricating grease not applied to packing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215340
4.4.2.9	Grenade Assembly, Prior to Loading Into Projectile				NEXT HIGHER ASSEMBLY 9362612
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety pin missing		100%	3.2	Visual
2	Slider lock cracked, incomplete, missing or broken		100%	3.2	Visual
3	Slider lock improperly positioned such that the slide would not be restrained		100%	3.2	Visual
Major					
101	Safety clip not removed		0.40%	3.2	Visual
102	Ribbon unwound		0.40%	3.2	Visual
103	Tape not completely inserted in slots of stiffener		0.40%	3.2	Visual
Minor					
201	None defined				
REMARKS					

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Projectile, After Removal of Safety Pins From First Layer of Grenades and Prior to Any Further Loading	1 OF 1	9362612
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Slide assembly, protruding from grenade fuze housing indicating armed or partially armed fuze		100%	3.2	Visual
2	Grenade missing or improperly assembled		100%	3.2	Visual
3	Slider lock broken, incomplete, or missing		100%	3.2	Visual
4	Spacer missing, large and small (wrong color), or improperly positioned		100%	3.2	Visual
* 5	Slider lock improperly positioned such that slider is not restrained		100%	3.2	Visual
Major					
101	Safety clip or safety pin not removed from grenade		100%	3.2	Visual
102	Any stiffener or tape incorrectly positioned or missing		100%	3.2	Visual
* 103	Booster PA 530 missing or reversed		100%	3.2	Visual
* 104	Improper mixing of solid and cored spacers		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.11	Projectile After Removal of Safety Pins From Layer and Prior to Any Further Loading Applicable to Second through Twelfth Layers Inclusive		1 OF 1	9362612
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Slide assembly, protruding from grenade fuze housing, indicating armed or partially armed fuze		100%	3.2	Visual
2	Grenade missing or improperly assembled		100%	3.2	Visual
3	Slider lock broken, incomplete, or missing		100%	3.2	Visual
4	Either type key missing or incorrect		100%	3.2	Visual
5	Center spacer missing		100%	3.2	Visual
* 6	Spacer missing, large and small, or improperly positioned		100%	3.2	Visual
7	Slider lock improperly positioned such that the slide would not be restrained		100%	3.2	Visual
Major					
101	Safety clip or safety pin not removed from grenade		100%	3.2	Visual
102	Stiffener or tape incorrectly		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

REMARKS

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Projectile and Base, After Assembly of Adapter and Prior to Assembly of Base and O-Ring	1 OF 1	9362612
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Adapter missing (15)		100%	3.2	Visual
2	Center spacer missing		100%	3.2	Visual
3	Either type key missing		100%	3.2	Visual
4	Spacer missing, large or small, or improperly positioned		100%	3.2	Visual
Special					
a	O-ring missing or wrong type		100%	3.2	Visual
b	Sealant missing on threads of base		100%	3.2	Visual
c	Threads damaged on base		100%	3.2	Visual
d	Threads damaged on base of projectile body		100%	3.2	Visual
Major					
101	Depth between projectile face and adapters or shims as applicable at specified load		100%	3.2	4.5.12
102	O-Ring not greased		0.40%	3.2	Visual
103	Foreign materials on threads		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

BUYER:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Projectile, Prior to Assembly of Expulsion Charge Assembly and Grommet	1 OF 1	9362612
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
* a	Gap between projectile body and base, 360°		100%	3.2	Gage
b	Disassembly torque test of base		100%	3.2	4.5.14
* c	Bare spots in anodic coating on base		100%	3.2	Visual
Major					
101	Depth of surface of projectile to expulsion charge cup, min		0.40%	3.2	Gage
102	Expulsion charge cup improperly staked		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.14	Projectile, Prior to Assembly of Fusible Lifting Plug		1 1		9362612
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Weight		100%	3.2	Gage
* B	Bare spots on protective coating		100%	3.2	Visual
Major					
101	Expulsion charge missing		0.40%	3.2	Visual
102	Nose threads damaged for length specified		0.40%	3.2	Visual
103	Notch in expulsion charge assembly damaged		0.40%	3.2	Visual
104	Zone weight marking missing, illegible or incorrect		100%	3.2	Visual
105	Thread in lifting plug damaged		0.40%	3.2	Visual
106	Diamond marking missing, incorrect or unidentifiable		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9362612
4.4.2.15	Projectile				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Lifting plug other than fusible type		100%	3.2	Visual
2	Air pressure test		100%	3.2	4.5.6
Special					
* a	Lifting plug or gasket missing or damaged		100%	3.2	Visual
Major					
101	Lifting plug fails to withstand the specified disassembly torque	(a)	0.40%	3.2	4.5.13
102	Obturator missing or damaged		0.40%	3.2	Visual
103	Bourrelet diameter, max (applicable only when projectile paint on bourrelet has been retouched)		100%	3.2	Gage
104	Radiographic examination	(b)		3.7	4.5.8
Minor					
201	Required marking missing, incorrect or unidentifiable		0.65%	3.2	Visual

NOTES:
(a) Sample size per MIL-STD-105, Inspection Level I.
(b) Ballistic samples only.

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9362612
4.4.2.15	Projectile				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
202	Rotating band damaged (applicable if grommet is missing or damaged)		0.65%	3.2	Visual
203	Grommet assembly missing, damaged or improperly assembled		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9229038
4.4.2.16	Pallet Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Special * a	Cover support missing (6 places)		100%	3.2	Visual
Major 101	Strapping loose or improperly assembled		0.40%	3.2	Visual/Manual
102	Seal strapping missing		0.40%	3.2	Visual
103	Lifting plug of projectile missing or damaged		0.40%	3.2	Visual
Minor 201	Board broken or split		0.65%	3.2	Visual
202	FSN number or descriptive nomenclature missing or unidentifiable		0.65%	3.2	Visual
203	Any other marking missing, incorrect or not identified		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Adapter, Prior to Assembling into Projectile	1 OF 1	9215380
			NEXT HIGHER ASSEMBLY 9362612

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Verification of heat treatment of adapter		100%	3.2	4.5.10
Major	None defined.				
Minor 201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC n 1570, 1 Feb 85 Replaces SMC-QA (D) Form 160, 1 Aug 83, which may not be used

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.18	Shim		1 1		9272006
					NEXT HIGHER ASSEMBLY 9362612
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Hardness		0.40%	3.2	4.5.9
102	Diameter, min.		0.40%	3.2	Gage
103	Thickness, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9378249
4.4.2.19	Pellet				NEXT HIGHER ASSEMBLY 9378247
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Crack or lamination		0.40%	3.2	Visual
102	Length (dwg. 9378249-1 or -2)		0.40%	3.2	Gage
103	Diameter		0.40%	3.2	Gage
104	Moisture content of explosive charge material		(a)	3.2	4.5.3
105	Density of explosive charge material		(b)	3.2	4.5.4
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) Sampling in accordance with 4.5.3
(b) Sampling in accordance with 4.5.4

AMSMC Form 1570, 1 Feb 85 Replac RSMC-QA (D) Form 160, 1 Aug 83, which may not be use

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9378250
4.4.2.20	Pellet				NEXT HIGHER ASSEMBLY 9378247
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9378248
4.4.2.21	Cup		NEXT HIGHER ASSEMBLY 9378247

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Length, internal		0.40%	3.2	Gage
102	Length, external, max.		0.40%	3.2	Gage
103	Thickness, base		0.40%	3.2	Gage
104	Thickness, wall		0.40%	3.2	Gage
105	Diameter, internal		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

REVERSE

AMSMC m 1570, 1 Feb 85 Replace SMC-QA (D) Form 160, 1 Aug 83, which may not be used

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9378251
4.4.2.22	Cup		NEXT HIGHER ASSEMBLY 9378247

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Thickness		0.40%	3.2	Gage
102	Diameter, inside		0.40%	3.2	Gage
103	Diameter, outside		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Obturator	1 OF 1	9287983
			NEXT HIGHER ASSEMBLY 9362612

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Inside diameter		0.40%	3.2	4.4.3.3/Gage
102	Thickness of band		0.40%	3.2	4.4.3.3/Gage
103	Thickness at small outside diameter		0.40%	3.2	4.4.3.3/Gage
104	Distance between two inside diameter		0.40%	3.2	4.4.3.3/Gage
105	Length of small inside diameter		0.40%	3.2	4.4.3.3/Gage
106	Breaking load test	*		3.2	4.4.15
107	Overall length		0.65%	3.2	4.4.3.3/Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	4.4.3.3/Visual

NOTES: *Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12576778
4.4.2.24	Lid, Container, Expulsion Charge		NEXT HIGHER ASSEMBLY 12576776

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter of locking lip		0.40%	3.2	Gage
103	Inside diameter of open end		0.40%	3.2	Gage
104	Inside diameter of closed end		0.40%	3.2	Gage
105	Length of cylindrical portion of lid		0.40%	3.2	Gage
106	Incomplete or missing tab			3.2	Visual
107	Profile of locking lip area (destructive)	(a)	0.40%	3.2	Gage
108	Total height of lid		0.40%	3.2	Gage
109	Depth of cap		0.40%	3.2	Gage
110	Thickness of tab below ridges		0.40%	3.2	Gage
111	Width of ribs		0.40%	3.2	Gage
112	Height of ribs		0.40%	3.2	Gage
113	Pull test, min.	(b)		3.2	4.5.16
Minor					
201	Missing slot (4 places)		0.65%	3.2	Visual
202	Presence of friction ribs on tab		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	4.4.3.3/Visual

NOTES:
(a) Sample three (3) parts from each mold cavity at the end of each shift.
(b) Sample ten (10) per lot: Accept no failures; Reject on 1 failure.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12576779
4.4.2.25	Cup, Container, Expulsion Charge		NEXT HIGHER ASSEMBLY 12576777

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Length of cup		0.40%	3.2	Gage
102	Length of cap interlock area		0.40%	3.2	Gage
103	Outside diameter of ribs		0.40%	3.2	Gage
104	Outside diameter of locking lips		0.40%	3.2	Gage
105	Inside diameter at top of cup		0.40%	3.2	Gage
106	Outside diameter at bottom of cup between ribs		0.40%	3.2	Gage
107	Thickness of bottom		0.40%	3.2	Gage
108	Profile of locking lip area		0.40%	3.2	Gage
109	Pull test, min.			3.2	Gage
110	Flatness on seal area		0.40%	3.2	Gage
111	Wall thickness variation between ribs from top to bottom of cup		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
(a) Sample ten (10) per lot: Accept no failures; Reject on 1 failure.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Disc, Expulsion Charge	1 OF 1	9362554
			NEXT HIGHER ASSEMBLY 12576777

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Thickness		0.40% (a)	3.2	Gage
102	Diameter		0.40% (a)	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) 0.40% AQL or alternatively, if the disc is used to seal the expulsion cup immediately after the disc is made, then take 5 samples per machine at the start of production each day and 5 samples per machine after every 4 hours of production or protion thereof. If any part fails to comply with the requirements, the parts represented by the sample since the last satisfactory inspection shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Container Assembly, Expulsion Charge	1 OF 1	12576777
			NEXT HIGHER ASSEMBLY 12576777

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Leak Test	(a)		3.2	4.5.2.1
102	Weight of propellant charge incorrect		0.40%	3.2	Gage
103	Disc seal incomplete		0.40%	3.2	Visual
104	Moisture and total volatiles content of propellant	(a)		3.2	4.5.1
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
(a) Sample in accordance with applicable test method paragrah.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48091G (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12576776
4.4.2.28	Expulsion Charge Assembly				NEXT HIGHER ASSEMBLY 9362612
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Cup or lid damaged		0.40%	3.2	Visual
102	Gap between cup and lid 360°		0.40%	3.2	Gage
103	Total length, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing. Testing is required as described herein and in the First Article and Quality Conformance Inspection Table.

4.4.3.1 Ballistic functioning initial production (major defect) (see 3.6). Beginning with the first lot produced and continuing until three (3) consecutive lots have been accepted, the Government inspector shall select ten (10) projectiles from each lot for this test. Rounds shall be tested as specified in 4.4.3.2.2.2. Rejection shall be as specified in 4.4.3.2.3.2.

4.4.3.2 Ballistic functioning, regular production (major defect).

4.4.3.2.1 Sampling - Following successful completion of the tests specified in 4.4.3.1 and with the approval of the Contracting Officer, the Government inspector shall select six (6) projectiles from each lot for this test. They shall be tested per 4.4.3.2.2.1. However, if any lot is ballistically rejected, as determined by the criteria in 4.4.3.2.3, increased sampling and testing shall be continuously applied until three (3) consecutive lots have been accepted in accordance with the criteria of 4.4.3.1. Thereafter, sampling and testing of subsequent lots shall be in accordance with 4.4.3.2, or as directed by the Contracting Officer.

4.4.3.2.2. Testing - The projectiles selected in 4.4.3 shall be forwarded to the Government proving ground designated by the Contracting Officer, for testing in accordance with the methods and procedures specified in 4.5.8. The projectiles shall be tested in accordance with the following plans, as applicable.

4.4.3.2.2.1 Sample size of six (6) projectiles - The six (6) projectiles selected from the applicable lot, shall be tested at the following charge zones:

a. Two (2) of the projectiles shall be tested at Zone 1. (See 4.5.7.1)

b. Two (2) of the projectiles shall be tested at Zone 9. (See 4.5.7.2)

c. One (1) of the projectiles shall be tested at Zone 9X. (See 4.5.7.3)

d. The remaining projectile shall be used for replacement or retest if required, (See 4.4.3.2.4 and 4.4.3.2.5). If the projectile is not required for this purpose it shall be fired at Zone 9X and the test results recorded and forwarded as specified in 4.5.7.4.

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4.4.3.2.2 Sample size of ten (10) projectiles. The ten (10) projectiles selected from the applicable lot, shall be tested at the following charge:

a. Three (3) of the projectiles shall be tested at Zone 1 (See 4.5.7.1)

b. Three (3) of the projectiles shall be tested at Zone 9 (See 4.5.7.2)

c. Three (3) of the projectiles shall be tested at Zone 9X (See 4.5.7.3)

d. The remaining projectile shall be used for replacement or retest if required, (See 4.4.3.2.4 or 4.4.3.2.5). If the projectile is not required for this purpose, it shall be tested at Zone 9X and the test results recorded and forwarded as specified in 4.5.7.4.

4.4.3.2.3 Rejection. The lot represented by the sample shall be rejected (see Note 3) if the ballistic test results for that lot reveal any one or more of the following defects:

a. Any premature burst occurs (See 6.6.1)

b. Any separation of metal parts occurs prior to initiation of the time fuze.

c. Any detonation of the projectile cargo occurs prior to initiation of the time fuze.

d. The total number of zone 1 and 9 grenades failing to satisfactorily function (see 6.7.1) equals or exceed thrity seven (37) for a sample size of six (6) projectiles and equals or exceeds fifty-five (55) for the sample size of ten (10) projectiles (not applicable to zone 9X tests).

e. Any projectile fails to function satisfactorily (See 6.6.2 and 6.6.3).

f. If the total number of air detonations from the entire four (4) projectiles tested in zones 1 and 9 exceed sixteen (16) for six (6) projectiles or exceeds twenty-four (24) for ten (10) projectiles the data shall be referred to the engineering agency (see 6.4) for decision on final acceptance of the lot.

NOTE 1: A round by round accounting shall be made for lost grenades and the replacement provisions utilized when applicable.

NOTE 2: Grenade air detonations (see 6.7.2) shall be counted and recorded for informational purposes.

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NOTE 3: If any lot is rejected, sampling and acceptance for ballistic functioning shall revert back to the criteria of 4.4.3.1.

4.4.3.2.3.1 Damaged grenades. Grenades evaluated to have malfunctioned due to damage resulting from air collision/detonations shall not be included in the total number of grenades used to accept or reject the lot. However, if the total number of such grenades from six (6) rounds tested in zone 1 and 9 exceeds 30 or from four (4) rounds tested in zone 1 and 9 exceeds 20 the data shall be referred to AMSMC-QAR-Q (D) for decision on final acceptance of the lot. Grenades so malfunctioning shall be counted and recorded on the evaluation sheet for informational purpose.

4.4.3.2.4 Projectile replacement. If any projectile, except those tested at Zone 9X safety phase, functions below eight hundred (800) feet and the total number of all grenade defects does not exceed the specified acceptance criteria, no replacement for that projectile shall be required. However, when the results of the low bursting projectile would cause the lot represented by the sample to be rejected, the results of that projectile shall be disregarded and a replacement projectile shall be fired in the same charge zone. The results of the replacement projectile shall be substituted.

Time fuzes malfunctions shall not be considered as cause for projectile rejection. When a time fuze malfunction occurs, a replacement projectile may be fired in the same charge zone.

A replacement projectile is required whenever more than twenty (20) grenades from one projectile (except those fired at Zone 9X) are considered to be lost. Results of the rounds with excessive lost items shall be recorded for informational purposes. The replacement shall be fired in the same charge zone and results of same shall be used in determination of lot acceptance.

4.4.3.2.5 Retest. If for any reason the proving ground considers that test conditions have affected all or part of the test results, the proving ground shall declare the affected results invalid, and preform new tests on additional projectiles, as required.

4.4.3.3 Dimensional control of molded parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see Note 1), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to quality a new or reworked cavity for use in production or as a control of each cavity during production.

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The inspection of the parts shall be performed only when the curing time for the parts shall be performed on the above quantities from each cavity after production of each 500 parts or at the end of the shift is less than 500 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective parts will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

The contractor shall furnish evidence to the Government inspector that dimensional stability has been attained prior to measuring the parts for acceptance.

NOTE 1: Establish a curing time as follows: Dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins). Curing time will be from the time the part comes our of the mold until dimensional stability is attained. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.4.4 Inspection equipment. For the inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of item produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

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4.4.4.2 NDT personnel qualification and certification requirements.

4.4.4.2.1 Radiographic inspection. For setting up and performing tests using x-ray equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I or higher. Level I personnel shall work under the technical supervision of Level II personnel.

For set-up and calibration of x-ray equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II. Personnel who read the interpret radio-graphic images and prepare radiographic inspection reports shall be certified to Level II or higher.

A person certified to Level II shall approve all radiographic procedures for exposure, interpretation, reporting and changes thereto and be available for resolution of radiographic problems.

4.4.4.2.2 Other inspections. Personnel validating or operating non-destructive test equipment shall be qualified and certified in accordance with MIL-STD-410.

4.5. Methods of inspection.

4.5.1 Moisture and total volatiles content of charge assembly propellant. The moisture and total volatiles content of each lot or propellant or portion thereof shall be determined for each eight (8) hours of production. This determination shall be accomplished prior to the anticipated use of the lot or portion thereof, with a maximum of eighteen (18) hours lead time. Sampling of the propellant shall be in accordance with MIL-STD-286, Method 101.1.2. If the moisture and total volatiles content fails to comply with the requirements of MIL-P-48099, the propellant lot shall be rejected until corrective action has been taken as verified by repeating the moisture and total volatiles content determination. The contractor shall provide adequate controls to insure maintaining compliance with the requirements between testing and use.

4.5.2 Leak test of explusion charge assembly (destructive test). A sample of five (5) assemblies from each hours production per sealing machine shall be selected for this test. Each sample assembly shall be subjected to the vacuum specified on the applicable drawing using equipment and procedures shown on

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drawings 8841732 and 9202990 (except test instruction 4, see Note 1 for reject criteria). If any sample assembly fails the leak test, (see Note 1) the quantity represented by that sample shall be set aside and retested per the following procedure:

a. Visually inspect all bags for completeness of heat seals (top, side, and bottom). Bags with damaged or incomplete seals shall be classed defective and removed from production.

b. From remaining assemblies, ten (10) shall be selected and subject to the leak test, reject on 1 or more defects.

4.5.2.1 Leak test of expulsion charge container assembly-dwg. 12576776 (destructive test). A sample of five (5) assemblies (without lids) from each hour's production per assembly shall be subjected to the vacuum specified on the applicable drawing using equipment and procedures shown on drawings 8841732 and 9202990 (except test instruction 4, see Note 1 for reject criteria). If any sample assembly fails the leak test (see Note 1), the quantity represented by that sample shall be set aside and retested per the following procedure:

a. Visually inspect all cups for completeness of heat seals. Cups with damaged or incomplete seals shall be classed defective and removed from production.

b. From remaining assemblies, ten (10) shall be selected and subjected to the leak test, accept on 0, reject on 1 defect.

NOTE 1: For the purpose of this test, a reject shall be defined as an assembly that leaks to the extent that the M10 propellant becomes wet. To determine the condition of the propellant those assemblies that exhibit bubbles during the test shall be cut open and the propellant examined (visually and manually) for moisture. Only those assemblies containing moisture shall be considered defective.

4.5.3 Moisture content of explosive charge material for pellet (dwg. 9378249). When the explosive materiel is suspected of having been subjected to adverse storage or shipping conditions permitting the possibility of moisture contamination, a determination of moisture content shall be made. A representative sample of each material shall be selected and tested in accordance with MIL-P-48395. If the moisture content fails to comply with the requirement specified in the applicable specification, and loading has not begun, the explosive represented by the sample shall not be used for production until corrective action has been taken as verified by repeating this test. In addition, if loading has begun, booster PA 530 assemblies with the non-conforming composition shall be rejected.

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4.5.4 Consolidation density of explosive composition of pellet (dwg. 9378249). The density of the explosive composition of the pellet shall be determined for each consolidate tool at the start of production each day and after every two hours of production or protion thereof. If the density of any pellet in a representative test sample of three (3) fails to comply with the applicable drawing requirement, the pellets products since the last satisfactory check shall be rejected. In addition, the necessary adjustments shall be made to the consolidate tool which will be verified by determining the density of three (3) consecutively produced pellets.

The density of the explosive composition of the pellet shall be calculated as follows:

$$\text{Density, gm/cc} = \frac{\text{Weight in gms}}{(\text{Height, inches})\ (\text{Diameter, inches})^2\ 12.870}$$

4.5.5 Static functioning test of Booster PA 530 (see dwg. 9378249). The samples to be subjected to this test shall be tested by any method satisfactory to the Contracting Officer, providing the following conditions are included in the method.

a. A spotting charge assembly (Ref. dwg. 9272016) standard for this projectile, from accepted lots of spotting charge assemblies shall be used.

b. The spotting charge assembly shall be initiated using an output equivalent to that of the fuze standard for this projectile (Reference Dwg. 9236500, 9236502, 9236504).

c. The cup loading assembly shall be surrounded by a ring of five (5) body loading assemblies, (Ref. dwg. 9215342), oriented with respect to each other and the cup loading assembly as in the projectile design. The cluster thus formed shall be maintained by a confining ring of any material.

d. A disc, of the same material specifications and thickness as the forward plate, (Ref. Dwg. 9210124), shall be set on the booster PA 530.

e. The orientation of the items in the cluster, the disc, and the stand-off distance, relative to the spotting charge assembly and fuze, shall be the same as in the projectile design (Destructive test).

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4.5.6 Air pressure test of projectile. The assembled projectile, with lifting plug, shall be placed in a chamber and a pre-determined volume of air shall be delivered to produce the specified pressure between the inside of the chamber and the outside of the projectile. If the indicated pressure remains constant for the specified time interval that projectile will be considered to have passed this test. HOwever, if the indicated pressure at the end of the time interval is less than the specified indicated pressure at the beginning of the time interval, that projectile shall be suitably identified and removed from the lot to prevent mixing with unsatisfactory projectiles. Equipment used to perform this test shall be in accordance with 4.4.4 and shall be calibrated on an approved periodic basis to confirm the accuracy of the test results.

4.5.7 Ballistic functioning tests. The following tests shall be performed at a Government-owned proving ground in accordance with Aberdeen Proving Ground Acceptace Test Procedure LO-VAR and Supplement LO-8H-3J. The round shall be fitted with an M577 fuze. The projectile fuze setting and quadrant elevation of the weapon shall be adjusted to attempt to achieve ejection, at the target height specified below, over a prepared impact area. The maximum ejection height shall be nineteen hundred (1900) feet, for any zone. A photograph shall be taken of the fuze setting for each round immediately prior to use and identified with the round number. The sample projectile shall be tested at the specified zones using weapons, equipment methods and procedures in accordance with the acceptance test procedures referenced above. Tubes used for this test shall not have less than twenty five (25) percent remaining tube life based on current gun tube condemnation criteria.

4.5.7.1 Zone 1. The sample projectiles to be subjected to this test shall be fired in the M201 Cannon, using the M1 propelling charge. The target ejection height for this phase shall be one thousand (1000 feet). Observation shall be made for failure to comply with the requirements of 3.6.

4.5.7.2 Zone 9 - The sample projectiles to be subjected to this test shall be fired in the M201 Cannon using the M188A1 propelling charge. The target ejection height for this phase shall be sixteen hundred (1600) feet. Observation shall be made for failure to comply with the requirements of 3.6.

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4.5.7.3 Zone 9X - Safety phase. The sample projectiles to be subjected to this test shall be fired, using a propelling charge a conditioned to +145^{0}F with propellant added as necessary to produce a pressure of 44,300 to 48,100 PSI. Observation shall be made for failure to comply with the requirements of 3.6.3. Burst height and grenade functioning are not considered as part of this test.

4.5.7.4 Records of ballistic tests. A record shall be made for each round fired and shall include an accounting for the defects listed in 4.3.3 or 4.4.3.2.3 as applicable. In addition, a record shall be made of fuze duds, fuzes with excessive off-times, and any other malfunctions. A copy of the records of all ballistic test results shall be forwarded to the address specified in 6.4.

4.5.8 Radiographic Inspection of Projectile The projectile assemblies shall be readiogrphically examined prior to forwarding to the proving ground. Prior to assembly of the base, the Government inspector shall indicate the location of the key (9210135-1) by marking the outside of the projectile. In all exposures the radiographic beam shall be focused along the center axis of the projectile and parallel to the face of the forward plate, with the key perpendicular to the direction of the beam. Radiographic exposure shall, as a minimum, include all the base material adjacent to the adapters to and including the entire forward plate. If necessary, more than one radiograph shall be taken in this view as required to avoid adverse parallax effects and to observe the conditions specified in 3.7.

In the event the radiogrphs taken in this view prevent a suitable examination, the projectile shall be radiographed at a location rotated 90^{0} from the original with a spline nearest the source. Projectiles and radiographs shall be suitably identified to permit cross reference. Radiographs shall be maintained by the contractor until final acceptance of the production lot.

NOTE: Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of radiographic techniques shall be capable of at least two (2) percent (e.g. radiographic quality level 2-2T and the photographic density of the film used shall be of a quality which will produce a sharp definiton of internal structure. If any defect per 3.7 is found it shall be reported to AMSMC-QAR-Q (D). The lot represented by this sample shall be rejected.

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4.5.9 Hardness test of shim (dwg. 9272006) (Non-destructive). This test shall be performed in accordance with ASTM E-18 using the standard penetrator for the specified hardness scale. Three readings shall be taken approximately 120^0 apart. The average of the three readings shall be computed and used as the hardness value for that particular shim. If any reading fails the requirement on the applicable drawings, that shim shall be classed defective.

4.5.10 Verification of heat treatment of adapter. Each adapter shall be tested in accordance with the methods and procedures specified in ASTM E 18. One reading shall be taken on any acceptable surface. Adapters failing to comply with the specified requirement shall be rejected and removed from the lot. The contractor may use any other non-destructive test method, such as eddy current testing providing approval of the method is obtained from the Contracting Officer. Prior to such approval, the contractor may be required to submit all requested test data and operating procedures as required for evaluation of the method. Equipment used for performing this test shall be in accordance with 4.4.4 (Non-destructive test).

4.5.11 Strength of heat seal. This test shall be performed in accordance with FED-STD-101, Method 2024, as modified herein. The test samples shall be conditioned prior to testing at a temperature of 70 to 80 degrees Fahrenheit (F) and shall be tested under these conditions unless otherwise specified. One specimen, one inch wide, shall be cut perpendicular to the line of a seam of each sample bag. Location of the sample shall be random so that both the end seam and the long seam are sampled. One leg of the specimen shall be clamped so that the specimen hands freely. The required load (2.5 lbs) shall be applied to the free end in such a manner as to avoid impact loading the seam. The load shall be allowed to act for a minimum of one minute after which the seam shall be examined for peel-back of the heat seal.

4.5.12 Depth to adapters or shims. This measurements shall be performed using automatic or semi-automatic inspection equipment. The design of this equipment shall be "fail-safe" to the extent that the projectile cannot be released for further processing until the proper number of shims have been inserted. Verification that the proper number of shims have inserted shall be accomplished by re-inspection of this depth.

4.5.13 Disassembly torque test of lifting plug assembly. The projectile with lifting plug assembled, shall be subjected to the disassembly torque specified on Dwg. 9362612. Any unit that fails to withstand the disassembly torque shall be classed defective.

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4.5.14 Disassembly torque test of base. The projectiles with base assembled, shall be subjected to the disassembly torque specified on the applicable drawing. Any unit that fails to withstand the disassembly torque shall be classed defective.

4.5.15 Obturator breaking load test. A sample of twenty (20) obturators shall be randomly selected from each lot of obturators. If any of the samples fail to meet the requirements of Dwg. 9287983, the lot shall be rejected. The obturator shall be tested for breaking load using an approved test fixture. The test fixture shall consist of two one-inch diameter opposing spindles machined to the internal profile of the obturator. The obturator shall be assembled and tested using a test speed of 2.00 inches per minute.

4.5.16 Pull test of empty plastic expulsion charge assembly (destructive test). After each sample of the lid (dwg. 12576778) and the empty cup (dwg. 12576779) have been assembled together for this inspection, the assembly shall then be subjected to this test by placing it in a suitable fixture which will restrain the cup in such a manner as to avoid any squeezing on the body of the cup. The load, in an axial direction, shall then be applied to the tab of the lid, and the force required to separate the lid from the cup shall be measured. If any assembly fails to meet the minimum force requirement on the applicable drawing (12576776) the production quantity, both cup and lid, represented by that sample shall be rejected. All equipment used to perform this test shall be in accordance with 4.4.4.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. The projectile shall be packed in accordance with Dwg. 9229038.

5.2 Marking. Marking shall be in accordance with Dwg. 9229038 and Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Shipping. When shipments of more than one lot are made at one time, such as in a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. Not applicable to this specification.

6.2 Ordering Data. See MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: AMSMC-QAR-I (D), ARDEC, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Submission of ballistic test data and data cards. In addition to the normal distribution of records, when the projectile is procured by the U.S. Army Armament Materiel and Readiness Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, USA ARDEC, ATTN: AMSMC-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000.

6.5 Data requirements. Data item descriptions required for submission under this specification are as follows:

Paragraph	Data requirements	Applicable D.I.
4.4.2.9*, 4.4.2.10* 4.4.2.11*, 4.4.2.12* 4.4.2.15*, 4.5.2, 4.5.6, 4.5.9, 4.5.10, 4.5.11, 4.5.12, 4.5.13, 4.5.14	Quality Inspection Defect Report	DI-R-1721
4.5.1, 4.5.3, 4.5.4, 4.5.15	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724
6.12	Procedure Report	* DI-T-5372 (Tailored)

* Applies to critical defects only

** Delete a, b, c, d of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

6.6 Projectile fuctioning.

6.6.1 Premature burst. A premature burst is considered to have occurred if the projectile functions in the weapon or if the time of flight is less than 1.9 seconds.

6.6.2 Functioning at all zones, exluding Zone 9X - Satisfactory functioning of the projectile when fired at any zone, except Zone 9X, is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation of the time fuze at a pre-determined time after firing, and shall also include the successful expulsion of the entire cargo of grenades.

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6.6.3 Functioning at Zone 9X. Satisfactory functioning of a projectile fired at Zone 9X is defined as successful delivery of the projectile to the target area with all projectile hardware remaining intact until initiation of the time fuze.

6.7 Grenade functioning.

6.7.1 Unsatisfactory functioning. Grenades failing to function satisfactorily are defined as those grenades which upon ground impact do not function high order, except those grenades evaluated to have been damaged by air collision/detonation.

6.7.2 Air detonation. A grenade air detonation is defined as functioning of the grenade in the air due to any cause, during the time interval after expulsion from the projectile and prior to impact with the target.

6.8. Barricades. Approved barricades and test procedures shall be used when examination or tests are conducted on components or assemblies containing an explosive component.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal US Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is not under the cognizance of ARDEC.

6.10 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to a periodic verification and calibration check at frequent intervals during normal production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by 4.4.4.

6.11 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.12 Alternate inspection procedures.

6.12.1 Justification for alternate inspection procedures. Submitted documentation must include sufficient information to indicate an economic benefit, and must also clearly identify that there is no reduction in product quality.

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6.12.2 Alternate inspection procedures. The alternate inspection procedures as a minimum must include:

a. Frequency (production interval) for validating equipment functioning.

b. Proposed method of validation to incude process and inspection drawings.

c. Description of material control procedures to be imposed during the production interval between validations (Note: Under these provisions final acceptance will be withheld from material satisfactory validation).

6.12.3 Approval of alternate inspection procedures. Proposed plans shall be submitted per 6.3 for approval.

6.13 Subject term (keyword) listing.

Ammunition
Artillery
Grenades
Military Specification
Munition
Projectile, 8 inch, HE: M509

6.14 Changes from previous issue. TheThe margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 1320-A990)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-48091G (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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INCH-POUND
MIL-P-48091G (AR)
AMENDMENT 1
24 APRIL 1991

MILITARY SPECIFICATION

PROJECTILE, 8 INCH: HE: M509A1
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-48091G (AR), dated 30 November 1988, and is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 6

4.3.1: Delete "Expulsion Charge Assembly 9332505 25".

PAGE 7

TABLE I: Delete "Expulsion Charge Assembly (9332505)
Examination for defects 25 -- 3.2 4.4.2.4".

PAGE 9

4.4.1.g: Delete "g. Expulsioin charge assemblies from one lot interfix number from one manufacturer."

i: Delete "from one interfix number".

j: Delete "from one interfix number".

k: Delete "from one interfix number".

PAGE 11

4.4.2.1: Delete in its entirety.

PAGE 12

4.4.2.2: Delete in its entirety.

PAGE 13

4.4.2.3: Delete in its entirety.

PAGE 14

4.4.2.4: Delete in its entirety.

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PAGE 21

4.4.2.11, Critical defect 6: Add "(wrong color)" after "small".

PAGE 22

4.4.2.12, Title: Add "Layer" after "Adapter"

Critical defect 4: Add "(wrong color)" after "small".

PAGE 24

4.4.2.14, Title: Change "Fusible" to "Universal".

Major category: Delete Major defect 103

Add Major defect 107 as follows:

"107 Preformed packing ring present and properly assembled on lifting plug 100% 3.2 Visual".

PAGE 25

4.4.2.15, Critical defect 1: Change "Fusible type" to "Universal Lifting Plug".

Special defect a: Delete "or gasket".

Special category: Add "b Bare spots on protective coating 100% 3.2 Visual"

PAGE 27

4.4.2.16, Special defect a: Delete.

PAGE 35

4.4.2.24, Minor defect 201: Delete.

PAGE 44

4.5.2: Delete in its entirety.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-AC24)

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NOTICE OF INACTIVATION FOR NEW DESIGN

INCH POUND

MIL-P-48091G (AR)
NOTICE 1
14 December 1995

MILITARY SPECIFICATION

PROJECTILE, 8 INCH: HE: M509A1 LOADING, ASSEMBLING AND PACKING

This notice should be filed in front of MIL-P-48091G (AR) dated 30 November 1988

MIL-P-48091G dated 30 November 1988 with Amendment 1, dated 24 April 1991 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity:
Army - AR

AMSC N/A FSC 1320

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